

ASU Macro Qual Module 5

Sequential, Time-0, Recursive Competitive Equilibria, and Planner/Efficient Equilibria

PhD Qual Lecture Notes Builder Packet

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1 Orientation

Purpose of this packet

This module teaches how to *formulate* macroeconomic equilibria under exam pressure. The target skill is not memorizing one model; it is knowing how to translate any environment into the right list of equilibrium objects, agent problems, laws of motion, consistency conditions, market-clearing conditions, and planner comparisons. The central exam weakness this packet attacks is setup: choosing states, distinguishing individual and aggregate states, writing multiple value functions when needed, and making distribution laws explicit.

How to use this module

Read Sections 2–3 first to understand the source base and exam plan. Then work through the template sections in order:

1. Sequential competitive equilibrium and time-0 equilibrium.
2. Planner problems, welfare theorems, and wedge comparisons.
3. Recursive competitive equilibrium and stationary RCE.
4. Distribution laws and consistency.
5. PS4-style examples: taste shocks, skilled/unskilled labor, home production, sector-specific human capital, and heterogeneous firms.

Then use the exercise bank as a timed drill. A realistic study block is two 3-hour sessions: one for definitions and templates, one for applications and answer sketches.

Exam-facing rule

For equilibrium-definition questions, graders reward completeness more than algebraic cleverness. The safest answer has five blocks: **objects**, **agent problems**, **laws of motion**, **consistency**, and **market clearing**. For planner/welfare questions, add **feasible set**, **planner objective**, and **wedge comparison**.

2 Source Inventory

This inventory records the sources used, their role in the packet, and extraction caveats. The source hierarchy follows the ASU Macro Manifest: past ASU qualifiers and current problem sets determine exam relevance; professor notes determine notation and framing; homeworks determine practice style; outside textbooks are used only to fill conceptual gaps.

Source	Type	Topics used	Extraction/relevance notes
GitHub RCE folder: Macroeconomics/RCE/	Repository folder	Confirms core RCE source set: Natalia notes, PS4, Galina HW6/HW7, and class notes on time-0/sequential/RCE.	Folder listing was accessible. It also contains PDFs and TeX files, so TeX sources were preferred when available.

Source	Type	Topics used	Extraction/relevance notes
MACRO_MANIFEST.md	Project manifest	Module 5 required coverage, source hierarchy, notation conventions, PS4 anchors, past-qual patterns.	High authority for packet structure and topic priorities.
SKILL.md	Skill instructions	Required workflow, packet structure, RCE requirements, distribution-law requirement, planner/welfare requirements, TeX formatting.	High authority for organization and quality gates.
Class 09-10 Time0 CE WF Seq CE.tex	Professor notes	Sequential CE, time-0 Arrow-Debreu CE, welfare theorem logic, planner-vs-market framing.	TeX source available. Some Mathpix-style conversion artifacts; used for framing and definitions, not copied verbatim.
Class 11 - SCE:RCE.tex	Professor notes	Generic recursive CE structure, state variable discipline, individual vs aggregate states, consistency of aggregate laws.	TeX source available. Central source for setup logic.
4_RCE.pdf	Professor lecture note	Representative-agent RCE with leisure, big- K /little- k trick, price functions, consistency, adjustment-cost/dynamic-firm extension.	PDF extraction usable. The key conceptual point is that prices must be represented recursively by aggregate states and laws of motion.
5_RCE_with_Heterogeneous Agents.pdf	Professor lecture note	Bewley/Huggett/Aiyagari-style RCE, explicit distribution laws, stationary RCE, aggregate shocks, heterogeneous firms.	PDF extraction usable. Central source for law-of-motion templates.
PS4-6.pdf	Current problem set	RCE with taste shocks and inelastic labor, skilled/unskilled workers, tradable vs nontradable home good, sector-specific human capital, heterogeneous firms with switching costs.	Highest current-course anchor for Module 5. Problems are definition-heavy and require explicit distribution laws.
HW6 - Galina.pdf and HW7 - Galina.pdf	Homework	Sequential CE with taxes, government budget constraints, planner problem, efficiency iff no capital tax and constant sales tax, sequential two-good production/trade, payroll-tax RCE, heterogeneous firms.	Strong practice source for wedges, government budgets, and CE-vs-planner comparisons.

Source	Type	Topics used	Extraction/relevance notes
Past ASU Macro Qual TeX files 2016–2024	Past exams	Pareto weights and time-0 AD, asset-pricing RCE definitions, growth with government/taxes, externalities and planner wedges, stationary search RCE, entrepreneur RCE with switching costs.	Local TeX files were inspected. Exact year/problem mapping is listed later; some mappings are approximate when the source is only structurally related.
Ljungqvist and Sargent, <i>Recursive Macroeconomic Theory</i>	Supplementary textbook	Recursive methods, RCE conceptual discipline, heterogeneous-agent equilibrium background.	Used as a conceptual reference, especially for recursive formulation. The ASU materials determine notation.

3 Detailed Blueprint

3.1 Module objective

By the end of this packet, you should be able to take any macro environment and write a complete equilibrium definition in sequential, time-0, or recursive form. You should also be able to state and solve the associated planner problem, compare the CE and planner FOCs, identify wedges, and decide whether the allocation is efficient, constrained efficient, or inefficient.

3.2 Prerequisites

You should already be comfortable with:

- Bellman equations, FOC/envelope/Euler derivations, and feasibility constraints.
- Static firm profit maximization under constant returns to scale and decreasing returns to scale.
- Basic measure notation: integrals over distributions and measurable sets.
- Welfare theorem logic at the level of complete markets versus missing markets.

3.3 Canonical models covered

1. Representative-agent growth model with leisure: SCE, planner, and RCE.
2. Time-0 Arrow-Debreu equilibrium and its relation to sequential trading.
3. Taxed competitive equilibrium with sales taxes and capital-income taxes.
4. Heterogeneous-agent RCE with idiosyncratic shocks and distribution laws.
5. Stationary RCE.
6. PS4-style RCE variants: taste shocks, skill types, home production, sector-specific human capital, and heterogeneous firms with switching costs.
7. Dynamic firm RCE: why a firm value function is needed when firms have endogenous state variables.

8. Pareto weights, efficiency, constrained efficiency, and CE-vs-planner wedge comparisons.

3.4 Core problem archetypes

Prompt wording	What to write first	Main danger	Exam response
“Define a sequential CE”	Prices and allocations as sequences.	Forgetting one market or government budget.	HH problem, firm problem, government budget, market clearing, feasibility.
“Define a time-0 CE”	Date/state contingent commodities and time-0 budget.	Mixing sequential prices with AD state prices.	State-price budget, firm PV maximization, markets for all date/state commodities.
“Define an RCE”	Individual state, aggregate state, value/policy/price functions.	Omitting off-equilibrium state dependence.	Bellman equation, firm problem, law of motion, consistency, market clearing.
“Write the law of motion for the distribution”	State space and policy.	Writing only $K' = \int g d\mu$ when the state is a whole distribution.	Measurable-set formula for μ' .
“Is aggregate K enough?”	Ask whether prices and future aggregate laws depend only on K .	Confusing price sufficiency with state sufficiency.	State criterion; if policies depend on distribution, need μ .
“Set up planner; is CE efficient?”	Feasible set and planner FOCs.	Declaring any tax/non-tradability inefficient.	Compare CE FOCs to planner FOCs; identify actual wedge.
“Multiple value functions”	List agents with dynamic states.	Giving a single value function to every model.	HH value if HH has state; firm value if firm has state; planner value for allocation problem.

3.5 Closed-form and proof catalog

This module is definition-heavy, so closed forms are mostly “structural shortcuts”: Pareto sharing under complete markets, constant-sales-tax neutrality, Cobb-Douglas price functions, finite-state distribution laws, sector/home-production cutoffs, and stationary fixed-point equations. Each is derived in Section 24.

3.6 Exercise plan

The exercise bank is organized into mechanical drills, proof/closed-form drills, past-qual applications, and trap/economist-claim questions. The answer sketches focus on setup and key equations, not long algebra.

4 Module Map

Subtopic	Why it matters	Exam/source appearances	Core technique	Trap
Sequential CE	The default equilibrium concept in production and tax problems.	Galina HW6/HW7; past tax and government problems.	Sequences of allocations/prices; HH and firm optimization; market clearing.	Missing government budget or dividends.
Time-0 AD CE	Useful for welfare theorems and complete-market equivalence.	Class 09–10; 2016 Pareto/AD patterns.	Date/state-contingent commodities and state prices.	Using spot prices without securities.
Planner problem	The benchmark for efficiency.	PS1 planner mapping; Galina tax problem; past externality questions.	Feasible set, planner FOCs, compare to CE.	Planner should not face market prices.
Welfare theorems	Determines when CE is Pareto efficient.	Class notes and past AD questions.	Complete markets, convexity, local nonsatiation, no externalities.	Claiming inefficiency from a price wedge without checking feasible set.
Representative-agent RCE	Baseline for big- K /little- k .	Natalia RCE notes.	$v(k, K)$, price functions, aggregate law $H(K)$.	Writing $v(k)$ with time-varying prices.
Heterogeneous-agent RCE	Current PS4 emphasis.	PS4 Q1–Q5; heterogeneous agent notes.	State distribution μ , law $H(\mu)$, integrals.	Writing only average capital.
Stationary RCE	Search and heterogeneous-agent stationary equilibria.	2020 search SRCE; Aiyagari notes.	Fixed point $\mu^* = H(\mu^*)$.	Thinking individuals are stationary. Only distribution is stationary.

Subtopic	Why it matters	Exam/source appearances	Core technique	Trap
Home production tradeability	Tests efficiency logic and labor allocation.	PS4 Q3.	Tradable cutoff $p\theta$ versus w ; nontradable MRS condition.	Saying nontradable automatically inefficient.
Sector-specific human capital	Tests state choice and distribution dynamics.	PS4 Q4.	Individual state (h_A, h_B) and aggregate distribution.	Forgetting human capital evolves by sector choice.
Heterogeneous firms and switching costs	Tests multiple value functions.	PS4 Q5; 2024 entrepreneur RCE.	Firm state, firm value, switching policy, distribution law.	Treating dynamic firm problem as static.

5 Learning Objectives

After completing this packet, you should be able to:

1. Define a sequential competitive equilibrium for a production economy with households, firms, government, taxes, and dividends.
2. Define a time-0 Arrow-Debreu equilibrium and explain when it is equivalent to a sequential-trading equilibrium.
3. Write a representative-agent planner problem and compare its FOCs to CE FOCs.
4. Define an RCE by listing value functions, policy functions, price functions, aggregate laws, consistency conditions, and market-clearing conditions.
5. Write a distribution law in measurable-set form and in finite-state matrix form.
6. Decide whether aggregate K is a sufficient state or whether the whole distribution μ is needed.
7. Define a stationary RCE and explain why individual states may move while the distribution is fixed.
8. Set up RCEs for the five PS4 environments: taste shocks, skill types, home production, sector-specific human capital, and heterogeneous firms with switching costs.
9. Identify wedges caused by capital taxes, changing sales taxes, labor taxes, externalities, missing markets, and nonconvexities.
10. Answer economist-claim questions with formal equations and intuition.

6 Notation and Equilibrium Vocabulary

6.1 Time, agents, and states

Use $t = 0, 1, 2, \dots$ and $\beta \in (0, 1)$. Individual state variables are denoted by x ; aggregate state variables by S ; distributions by μ . In representative-agent economies, the aggregate state is often just aggregate capital K . In heterogeneous-agent economies, the aggregate state is usually a distribution,

for example $\mu(k, \theta)$ or $\mu(e, k)$.

6.2 Prices and timing

The final good is usually the numeraire. Factor prices are w_t for labor and r_t for rented capital. In recursive notation, prices become functions of the aggregate state: $w(S)$ and $r(S)$. A common gross return from holding one unit of capital from t to $t + 1$ is

$$R_{t+1}^k = 1 - \delta + r_{t+1}.$$

If capital income is taxed at rate τ_{t+1} , the after-tax gross return is

$$\tilde{R}_{t+1}^k = 1 - \delta + (1 - \tau_{t+1})r_{t+1}.$$

6.3 The five equilibrium blocks

Universal equilibrium-definition checklist

For almost every ASU macro equilibrium-definition question, write:

1. **Objects:** allocations, prices, value functions, policies, laws of motion, distributions.
2. **Household problem:** objective, state, controls, budget/feasibility, taking prices/laws as given.
3. **Firm problem:** static or dynamic profit/value maximization.
4. **Government rule:** taxes, transfers, debt, budget constraint, exogenous spending if present.
5. **Consistency and market clearing:** individual policies aggregate to aggregate laws; goods, labor, capital, asset, and goods-by-sector markets clear.

7 Sequential Competitive Equilibrium

7.1 Representative production economy with leisure

Consider a deterministic infinite-horizon economy with a representative household and representative firm. The household has preferences

$$\sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t),$$

where $c_t \geq 0$, $n_t \in [0, 1]$, and $1 - n_t$ is leisure. The firm has CRS production $F(K_t, N_t)$ and capital depreciates at rate δ .

7.2 Sequential CE definition

A **sequential competitive equilibrium** given $K_0 > 0$ is a sequence of allocations

$$\{c_t, n_t, k_{t+1}, K_t, N_t\}_{t=0}^{\infty}$$

and prices

$$\{w_t, r_t, \Pi_t\}_{t=0}^{\infty}$$

such that:

1. Household optimization. Taking prices and dividends as given, the household solves

$$\begin{aligned} \max_{\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t) \\ \text{s.t.} \quad & c_t + k_{t+1} \leq w_t n_t + r_t k_t + \Pi_t + (1 - \delta)k_t, \quad t \geq 0, \\ & c_t \geq 0, \quad k_{t+1} \geq 0, \quad n_t \in [0, 1], \quad k_0 = K_0. \end{aligned}$$

2. Firm optimization. For each t , the firm solves

$$\max_{K_t, N_t \geq 0} F(K_t, N_t) - r_t K_t - w_t N_t.$$

If F is differentiable and the solution is interior,

$$r_t = F_K(K_t, N_t), \quad w_t = F_N(K_t, N_t).$$

With CRS and competitive factor pricing, profits are zero: $\Pi_t = 0$.

3. Market clearing.

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t, \\ c_t + k_{t+1} - (1 - \delta)K_t &= F(K_t, N_t). \end{aligned}$$

The goods market clearing condition follows from the household budget, firm zero profit, and factor-market clearing, but write it explicitly unless the problem says “goods clears by Walras’ law.”

7.3 Sequential CE first-order conditions

Let λ_t be the multiplier on the household budget constraint. Interior choices imply

$$\begin{aligned} U_c(c_t, 1 - n_t) &= \lambda_t, \\ U_\ell(c_t, 1 - n_t) &= \lambda_t w_t, \\ \lambda_t &= \beta \lambda_{t+1} (1 - \delta + r_{t+1}). \end{aligned}$$

Therefore

$$\begin{aligned} \frac{U_\ell(c_t, 1 - n_t)}{U_c(c_t, 1 - n_t)} &= w_t, \\ U_c(c_t, 1 - n_t) &= \beta U_c(c_{t+1}, 1 - n_{t+1}) (1 - \delta + r_{t+1}). \end{aligned}$$

Substituting firm prices gives the equilibrium Euler and labor-leisure conditions:

$$\begin{aligned} \frac{U_\ell(c_t, 1 - n_t)}{U_c(c_t, 1 - n_t)} &= F_N(K_t, N_t), \\ U_c(c_t, 1 - n_t) &= \beta U_c(c_{t+1}, 1 - n_{t+1}) (1 - \delta + F_K(K_{t+1}, N_{t+1})). \end{aligned}$$

Economic intuition

The intratemporal condition says one more hour of work costs one unit of leisure and buys w_t units of consumption. The Euler equation says one unit of consumption sacrificed today buys one more unit of capital, which becomes $1 - \delta + r_{t+1}$ goods tomorrow.

Qualifier trap

The rental price r_t is the marginal product of capital rented *in period t* . The Euler equation uses r_{t+1} because choosing k_{t+1} today increases capital rented next period.

8 Time-0 Arrow-Debreu Equilibrium

8.1 Why time-0 CE is different

A time-0 Arrow-Debreu equilibrium opens all markets at date 0 for date-contingent and state-contingent commodities. A sequential CE opens spot markets over time and, if uncertainty is present, may also need securities markets to replicate Arrow-Debreu allocations. Time-0 CE is especially useful for welfare theorems and complete-market Pareto problems.

8.2 Deterministic time-0 budget representation

In a deterministic production economy, let $Q_0 = 1$ and define the Arrow-Debreu price of one date- t good in date-0 units by

$$Q_t = \prod_{s=1}^t \frac{1}{1 - \delta + r_s}, \quad t \geq 1,$$

when the sequential interest return is well-defined. Iterating the sequential budget gives the present-value budget constraint

$$\sum_{t=0}^{\infty} Q_t c_t \leq (1 - \delta + r_0)k_0 + \sum_{t=0}^{\infty} Q_t (w_t n_t + \Pi_t) - \lim_{T \rightarrow \infty} Q_T k_{T+1}.$$

Under the usual no-Ponzi or transversality condition, the limit term is nonnegative and equals zero at an optimum when the household does not throw away wealth.

8.3 Time-0 CE template

A **time-0 Arrow-Debreu competitive equilibrium** consists of allocations $\{c_t, n_t, k_{t+1}, K_t, N_t\}_{t \geq 0}$ and date-0 prices $\{Q_t, w_t, r_t, \Pi_t\}_{t \geq 0}$ such that:

1. Households choose entire sequences to maximize lifetime utility subject to a single present-value budget constraint.
2. Firms choose entire production plans to maximize present value of profits.
3. Government, if present, satisfies a present-value budget or state-by-state feasibility rule.
4. All date- t goods, labor, capital, and asset markets clear for every date and state.

Exam-facing rule

If the prompt says “Arrow-Debreu,” do not write only spot prices $\{w_t, r_t\}$ and call it done. You need commodities indexed by date and, under uncertainty, by history or state. If the prompt says “sequential trade,” do not collapse everything into one present-value budget unless you also specify the sequential asset structure.

8.4 Uncertainty and histories

If shocks follow histories $s^t = (s_0, \dots, s_t)$, then commodities are indexed by (t, s^t) . A time-0 household chooses $c_t(s^t)$, $n_t(s^t)$, and assets/capital plans for every history. State prices $Q_t(s^t)$ price goods delivered at history s^t . The time-0 budget takes the form

$$\sum_{t=0}^{\infty} \sum_{s^t} Q_t(s^t) c_t(s^t) \leq \text{present value of endowments, labor income, capital income, dividends, and transfers.}$$

With complete markets and convexity, time-0 CE and sequential CE with a complete set of Arrow securities implement the same allocations.

9 Planner Problems, Pareto Efficiency, and Welfare Theorems

9.1 Representative-agent planner

For the representative growth model, the social planner solves

$$\begin{aligned} V(K_0) = \max_{\{c_t, n_t, K_{t+1}\}_{t=0}^{\infty}} & \sum_{t=0}^{\infty} \beta^t U(c_t, 1 - n_t) \\ \text{s.t.} & \quad c_t + K_{t+1} \leq F(K_t, n_t) + (1 - \delta)K_t, \\ & \quad c_t \geq 0, \quad K_{t+1} \geq 0, \quad n_t \in [0, 1], \quad K_0 \text{ given.} \end{aligned}$$

The recursive form is

$$V(K) = \max_{K' \geq 0, n \in [0, 1]} \{U(F(K, n) + (1 - \delta)K - K', 1 - n) + \beta V(K')\}.$$

Interior FOCs imply

$$\begin{aligned} \frac{U_\ell(c_t, 1 - n_t)}{U_c(c_t, 1 - n_t)} &= F_N(K_t, n_t), \\ U_c(c_t, 1 - n_t) &= \beta U_c(c_{t+1}, 1 - n_{t+1}) (1 - \delta + F_K(K_{t+1}, n_{t+1})). \end{aligned}$$

These are the same as the CE conditions in the frictionless representative-agent economy.

9.2 Pareto planner with heterogeneous agents

Let agents be indexed by $i \in \{1, \dots, I\}$ and let Pareto weights be $\omega_i > 0$ with $\sum_i \omega_i = 1$. A complete-markets Pareto problem is

$$\begin{aligned} \max_{\{c_{it}, n_{it}, K_{t+1}\}} & \sum_{i=1}^I \omega_i \sum_{t=0}^{\infty} \beta^t U_i(c_{it}, 1 - n_{it}) \\ \text{s.t.} & \quad \sum_i c_{it} + K_{t+1} \leq F(K_t, N_t) + (1 - \delta)K_t, \\ & \quad N_t = \sum_i e_i n_{it}, \quad c_{it} \geq 0, \quad n_{it} \in [0, 1]. \end{aligned}$$

The intratemporal cross-sectional FOCs have the form

$$\omega_i U_{i,c}(c_{it}, 1 - n_{it}) = \lambda_t,$$

for consumption goods that all agents can trade. With log utility $U_i(c) = \log c$, this implies

$$c_{it} = \frac{\omega_i}{\sum_j \omega_j} C_t = \omega_i C_t.$$

Thus complete markets turn Pareto weights into consumption shares.

9.3 Efficiency concepts

Pareto efficient. No feasible reallocation can make one agent better off without making another worse off.

Constrained efficient. No feasible reallocation within the same constraints—for example, incomplete markets, nontradability, borrowing constraints—can improve welfare.

Inefficient. There exists a feasible reallocation, under the relevant feasible set, that improves welfare.

Qualifier trap

Do not say “nontradable good” or “tax” automatically means inefficiency. If nontradability is a technological or institutional feasibility constraint faced by the planner too, then the competitive equilibrium may be constrained efficient. If a tax is lump-sum, or a proportional consumption tax is constant over time and leisure is absent, it may create no real allocation wedge.

9.4 Welfare theorem logic

The First Welfare Theorem says that under complete markets, convex feasible sets, local nonsatiation, price-taking behavior, and no externalities/missing markets, competitive equilibrium allocations are Pareto efficient. The Second Welfare Theorem says that, under stronger convexity and transfer assumptions, Pareto efficient allocations can be decentralized by suitable wealth transfers and prices.

For exam use, the theorem is not a slogan. You must check whether the hypotheses are plausible:

- Missing insurance markets: First Welfare Theorem may fail relative to full-information allocations, but a CE can still be constrained efficient.
- Externalities: CE generally fails because agents/firms do not internalize aggregate effects.
- Taxes: efficiency depends on whether taxes create intertemporal or intratemporal wedges.
- Nonconvexities or switching costs: equilibrium may require dynamic firm values and may not be welfare-theorem-clean.

10 Recursive Competitive Equilibrium: General Template

10.1 Why recursive CE is not just a Bellman equation

A household Bellman equation takes prices as given. In equilibrium, prices are endogenous and usually change over time. RCE solves this by making prices functions of an aggregate state S and by specifying a law of motion $S' = \Lambda(S)$. Agents take S , price functions, and laws of motion as given; in equilibrium, the laws are consistent with their choices.

Exam-facing rule

The aggregate state has one job: it must contain enough information to determine current prices and agents' beliefs about future prices. If two economies have the same proposed aggregate state but different current prices or different future laws, the state is not sufficient.

10.2 Generic RCE environment

Let an individual have state $x \in X$, aggregate state $S \in \mathcal{S}$, and controls $a \in A(x, S)$. Let prices be $p(S)$, and let exogenous shocks follow transition Q . Let the individual policy be $g(x, S)$ and the aggregate law be $S' = \Lambda(S)$.

A **recursive competitive equilibrium** is a collection

$$(v, g, p, \Lambda, \text{firm policies, government rules})$$

such that:

1. Individual optimization. Taking $p(S)$ and $\Lambda(S)$ as given,

$$v(x, S) = \max_{a \in A(x, S)} \left\{ u(x, a, p(S)) + \beta \mathbb{E} [v(x', \Lambda(S)) \mid x, a, S] \right\},$$

and $g(x, S)$ is an optimal policy.

2. Firm optimization. For each aggregate state, firms solve their static or dynamic problems taking prices and laws as given. If firms have no endogenous state, this is usually static. If firms have capital adjustment costs, sector-switching costs, or productivity persistence, the firm has its own value function.

3. Government feasibility. Taxes, transfers, debt, and exogenous spending satisfy the government budget constraint state by state.

4. Consistency. The aggregate law Λ is generated by individual and firm policies. If the aggregate state includes K , then K' must equal aggregated individual saving. If the aggregate state includes a distribution μ , then $\mu' = H(\mu)$ must be the distribution induced by policies and shock transitions.

5. Market clearing. All markets clear for every feasible aggregate state S , not only along the equilibrium path.

11 Representative-Agent RCE: Big- K /Little- k

11.1 Objects

For the representative growth model with leisure, an RCE is a list of functions

$$(v, g_k, g_n, K', k^f, n^f, r, w),$$

where

- $v(k, K)$ is the household value function;

- $g_k(k, K)$ is individual next-period capital;
- $g_n(k, K)$ is individual labor;
- $K'(K)$ is the aggregate law of motion for capital;
- $k^f(K)$ and $n^f(K)$ are firm demands;
- $r(K)$ and $w(K)$ are price functions.

11.2 Household problem

Taking $r(K)$, $w(K)$, and $K'(K)$ as given,

$$v(k, K) = \max_{k' \geq 0, n \in [0,1]} U(w(K)n + r(K)k + (1 - \delta)k - k', 1 - n) + \beta v(k', K'(K))$$

s.t. $w(K)n + r(K)k + (1 - \delta)k - k' \geq 0$.

The policy functions $g_k(k, K)$ and $g_n(k, K)$ solve this problem.

11.3 Firm problem

For each K ,

$$(k^f(K), n^f(K)) \in \arg \max_{k, n \geq 0} \{F(k, n) - r(K)k - w(K)n\}.$$

Interior conditions are

$$r(K) = F_K(k^f(K), n^f(K)), \quad w(K) = F_N(k^f(K), n^f(K)).$$

11.4 Consistency and market clearing

In the representative-agent equilibrium, the individual's capital equals aggregate capital:

$$k = K.$$

The law of motion is consistent if

$$K'(K) = g_k(K, K).$$

Markets clear if

$$K = k^f(K), \quad g_n(K, K) = n^f(K).$$

Goods market clearing follows from the household budget, firm problem, and factor-market clearing.

Qualifier trap

Do not write $v(k)$ when prices vary over time. The household must know how future prices change. The aggregate state K is what makes current and future prices recursive.

11.5 Interpreting off-equilibrium states

An RCE defines policies for all pairs (k, K) , not only for $k = K$. This matters because the household is atomistic. It takes aggregate K as given and asks, “If I personally had k while the economy had aggregate K , what would I do?” Equilibrium imposes $k = K$ only in the consistency block.

12 Explicit Laws of Motion for Distributions

12.1 Measurable-set law

Let the individual state be $x \in X$, the aggregate state be a distribution $\mu \in \mathcal{P}(X)$, and the policy be $g(x, \varepsilon', S)$, where ε' is next period's idiosyncratic shock. For every Borel set $B \subseteq X$, the next distribution is

$$\mu'(B) = \int_X \int_{\mathcal{E}} \mathbf{1}\{g(x, \varepsilon', S) \in B\} Q(d\varepsilon' \mid x, S) \mu(dx).$$

If the current individual state already includes the shock, for example $x = (a, y)$, and next assets are deterministic $a' = g(a, y, S)$ while y' follows $Q(y, dy')$, then for a set $B = A \times Y$,

$$\mu'(A \times Y) = \int \mathbf{1}\{g(a, y, S) \in A\} Q(y, Y) \mu(da, dy).$$

12.2 Discrete-state law

Suppose $y \in \{\bar{y}_1, \dots, \bar{y}_N\}$ and $a \in \{\bar{a}_1, \dots, \bar{a}_M\}$. Let λ_{ij} be the mass of households with $y = \bar{y}_i$ and $a = \bar{a}_j$. If $g(\bar{a}_j, \bar{y}_i, \lambda) = \bar{a}_m$, then

$$\lambda'_{\ell m} = \sum_{i=1}^N \sum_{j=1}^M \mathbf{1}\{g(\bar{a}_j, \bar{y}_i, \lambda) = \bar{a}_m\} Q_{i\ell} \lambda_{ij}.$$

Equivalently, after ordering all states, $\lambda' = B(g, Q)\lambda$.

Exam-facing rule

When the prompt says “write an explicit law of motion for the probability distribution,” do not write only $K' = \int g d\mu$. That is the law for the aggregate mean. You need the push-forward formula for the whole distribution.

12.3 When is average K enough?

Average capital is

$$K(\mu) = \int k \mu(dk, d\theta).$$

Prices may depend only on $K(\mu)$. For example, with Cobb-Douglas production and inelastic labor $N = 1$,

$$r(\mu) = \alpha K(\mu)^{\alpha-1}, \quad w(\mu) = (1 - \alpha) K(\mu)^\alpha.$$

But price sufficiency is not state sufficiency. The aggregate law for next period capital is

$$K' = \int g_k(k, \theta, \mu) \mu(dk, d\theta).$$

If two distributions μ_1 and μ_2 have the same mean K but different masses of high-taste-shock households, and if those shocks change saving, then the two integrals can differ. Then K alone is not enough to determine future prices.

Criterion for reducing the aggregate state to K

You may use K as the only aggregate state only if both conditions hold:

1. Current prices are functions of K alone.
2. The induced aggregate law $K' = \int g_k(x, K) d\mu(x)$ is the same for every distribution μ with mean K ; equivalently, $K' = H_K(K)$ is well-defined.

In heterogeneous-agent incomplete-market models, condition 2 usually fails unless the problem has special aggregation.

13 Stationary Recursive Competitive Equilibrium

A **stationary RCE** is an RCE plus an invariant aggregate state. If the aggregate state is a distribution μ , stationarity requires

$$\mu^* = H(\mu^*).$$

If the aggregate state is (K, μ, z) and z is fixed in steady state, stationarity requires

$$K^* = K'(K^*, \mu^*, z^*), \quad \mu^* = H(K^*, \mu^*, z^*).$$

Economic intuition

In a stationary heterogeneous-agent equilibrium, individual households can move across states every period. A poor household can become rich, an unemployed worker can become employed, and a firm can switch sectors. Stationarity means the cross-sectional distribution is unchanged, not that each individual is unchanged.

13.1 Stationary RCE template

A stationary RCE for a Bewley-style economy is a list

$$(v, g, r, w, \mu^*)$$

such that:

1. Given constant prices r, w, v solves the household Bellman equation and g is optimal.
2. The distribution μ^* is invariant under g and the exogenous transition matrix.
3. Markets clear under μ^* .

For example, if assets must clear in zero net supply,

$$\int g(a, y) \mu^*(da, dy) = 0.$$

If capital is in positive supply and produced, market clearing is instead

$$\int k \mu^*(dk, de) = K^f(r, w).$$

14 Taxed Sequential CE and Planner Wedges

14.1 Environment

A representative household owns capital and firms. The government purchases an exogenous sequence $\{G_t\}_{t \geq 0}$ and finances it with lump-sum taxes T_t , a proportional sales tax s_t on consumption, and a proportional tax τ_t on capital rental income. The government budget is

$$G_t = T_t + s_t c_t + \tau_t r_t K_t.$$

To isolate the tax wedges, suppose utility is $u(c)$ and labor is supplied inelastically at $N_t = 1$. The household budget is

$$(1 + s_t)c_t + k_{t+1} \leq w_t + (1 - \tau_t)r_t k_t + (1 - \delta)k_t + \Pi_t - T_t.$$

14.2 Household FOCs

Let λ_t be the multiplier on the budget constraint. Interior conditions are

$$\begin{aligned} u'(c_t) &= \lambda_t(1 + s_t), \\ \lambda_t &= \beta \lambda_{t+1} [1 - \delta + (1 - \tau_{t+1})r_{t+1}]. \end{aligned}$$

Eliminate λ_t :

$$\frac{u'(c_t)}{1 + s_t} = \beta \frac{u'(c_{t+1})}{1 + s_{t+1}} [1 - \delta + (1 - \tau_{t+1})r_{t+1}].$$

Equivalently,

$$u'(c_t) = \beta u'(c_{t+1}) \frac{1 + s_t}{1 + s_{t+1}} [1 - \delta + (1 - \tau_{t+1})r_{t+1}].$$

14.3 Planner problem

The planner takes government purchases as resource costs:

$$\begin{aligned} \max_{\{c_t, K_{t+1}\}} \quad & \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t.} \quad & c_t + K_{t+1} + G_t \leq F(K_t, 1) + (1 - \delta)K_t. \end{aligned}$$

The planner Euler equation is

$$u'(c_t) = \beta u'(c_{t+1}) [1 - \delta + F_K(K_{t+1}, 1)].$$

In CE, $r_{t+1} = F_K(K_{t+1}, 1)$. Therefore the CE allocation satisfies the planner Euler equation if and only if

$$\frac{1 + s_t}{1 + s_{t+1}} [1 - \delta + (1 - \tau_{t+1})F_K(K_{t+1}, 1)] = 1 - \delta + F_K(K_{t+1}, 1)$$

for all t . A sufficient and, under nondegenerate capital returns, necessary condition is

$$\tau_t = 0 \quad \text{and} \quad s_t = s_{t+1} \quad \text{for all } t.$$

Economic intuition

A capital tax distorts the intertemporal margin by lowering the private return to saving. A changing sales tax distorts the intertemporal margin by making consumption today expensive relative to consumption tomorrow or vice versa. A constant sales tax multiplies all dates' consumption prices by the same factor and creates no intertemporal wedge when leisure is absent.

14.4 When would any positive consumption tax be distortionary?

Add leisure to utility, $U(c, 1 - n)$. With a sales tax, the intratemporal FOC becomes

$$\frac{U_\ell(c_t, 1 - n_t)}{U_c(c_t, 1 - n_t)} = \frac{w_t}{1 + s_t}.$$

The planner condition is

$$\frac{U_\ell(c_t, 1 - n_t)}{U_c(c_t, 1 - n_t)} = F_N(K_t, n_t) = w_t.$$

Thus any positive sales tax distorts the labor-leisure margin unless offset by another tax/subsidy.

Qualifier trap

If utility in the source problem is $u(c)$ only, do not invent a labor-leisure FOC. Labor is supplied inelastically or at a corner. If the question asks for a modification where consumption taxes distort, adding leisure to utility is the standard answer.

15 Multiple Value Functions: When One Bellman Is Not Enough

15.1 Household value versus firm value versus planner value

Use separate value functions when separate agents have separate dynamic states.

- A **household value function** is needed when households choose intertemporal saving, sector-specific human capital, employment/search, or other dynamic choices.
- A **firm value function** is needed when firms have a state variable that affects future profits: previous sector, installed capital, productivity persistence, adjustment costs, entry/exit status, or customer capital.
- A **planner value function** is not part of the CE definition. It is a separate allocation problem used for welfare and efficiency comparisons.

15.2 Dynamic firm template

Let a firm have state m and aggregate state S . It chooses a^f and earns current profit $\pi(m, a^f, S)$. If the firm discounts at price $q(S)$ or factor β_f , its Bellman equation is

$$V^f(m, S) = \max_{a^f \in A^f(m, S)} \left\{ \pi(m, a^f, S) + q(S) \mathbb{E}[V^f(m', S') \mid m, a^f, S] \right\}.$$

Examples:

- Capital adjustment costs: m may include last period's capital.

- Sector switching costs: m includes the previous sector.
- Persistent productivity: m includes current productivity.

Qualifier trap

If a firm's choice today affects its feasible set or cost tomorrow, the firm problem is dynamic. Do not write only $\max_n z f(n) - wn$ when the firm pays a switching cost based on last period's sector.

16 PS4 Template 1: Inelastic Labor and Idiosyncratic Taste Shocks

16.1 Environment

There is a continuum of households of mass one. Household taste shock θ follows Markov transition $Q(\theta, d\theta')$, with realizations iid across households. Preferences are

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t \theta_t u(c_t).$$

Labor is inelastic, so aggregate labor is $N = 1$. Production is Cobb-Douglas:

$$Y = K^\alpha N^{1-\alpha} = K^\alpha.$$

16.2 State variables

Individual state:

$$x = (k, \theta).$$

Aggregate state:

$$\mu \in \mathcal{P}(\mathbb{R}_+ \times \Theta).$$

Aggregate capital is the mean

$$K(\mu) = \int k \mu(dk, d\theta).$$

16.3 RCE objects

An RCE is functions

$$(v, g, c, r, w, H)$$

with

- $v(k, \theta, \mu)$: household value;
- $g(k, \theta, \mu)$: next capital;
- $c(k, \theta, \mu)$: consumption;
- $r(\mu), w(\mu)$: price functions;
- $H(\mu)$: distribution law.

16.4 Household problem

Taking prices and H as given,

$$\begin{aligned} v(k, \theta, \mu) = \max_{k' \geq 0} \quad & \theta u(c) + \beta \int v(k', \theta', H(\mu)) Q(\theta, d\theta') \\ \text{s.t.} \quad & c = w(\mu) + r(\mu)k + (1 - \delta)k - k', \\ & c \geq 0. \end{aligned}$$

16.5 Prices

Because labor is one,

$$r(\mu) = \alpha K(\mu)^{\alpha-1}, \quad w(\mu) = (1 - \alpha)K(\mu)^\alpha.$$

Thus the rental rate depends only on aggregate capital $K(\mu)$.

16.6 Distribution law

For any Borel sets $A \subseteq \mathbb{R}_+$ and $B \subseteq \Theta$,

$$H(\mu)(A \times B) = \int \mathbf{1}\{g(k, \theta, \mu) \in A\} Q(\theta, B) \mu(dk, d\theta).$$

16.7 Market clearing

Goods and capital markets clear if

$$\begin{aligned} \int [c(k, \theta, \mu) + g(k, \theta, \mu) - (1 - \delta)k] \mu(dk, d\theta) &= K(\mu)^\alpha, \\ K(H(\mu)) &= \int g(k, \theta, \mu) \mu(dk, d\theta). \end{aligned}$$

16.8 Can K be the only aggregate state?

The rental rate and wage depend only on K . But the household's future value depends on the future distribution if future prices or future policies depend on the distribution. Unless the problem gives or you prove an aggregate law $K' = \Gamma(K)$ independent of the cross-sectional distribution, K alone is not enough. The safe answer is:

Prices depend on K , but the RCE generally requires μ .

17 PS4 Template 2: Skilled and Unskilled Workers

17.1 Environment

There are two fixed types, skilled s and unskilled u . The mass of skilled workers is η and the mass of unskilled workers is $1 - \eta$. Type i households have utility

$$\sum_{t=0}^{\infty} \beta^t U(c_{it}, 1 - n_{it}).$$

The firm technology is CRS:

$$F(K, N_s, N_u).$$

17.2 State choice

If all households within each type remain identical and start with the same capital, a finite aggregate state such as (K_s, K_u) is enough. But the robust RCE definition uses distributions:

$$\mu_s(k), \quad \mu_u(k).$$

Then

$$\begin{aligned} K &= \eta \int k \mu_s(dk) + (1 - \eta) \int k \mu_u(dk), \\ N_s &= \eta \int n_s(k, \mu_s, \mu_u) \mu_s(dk), \\ N_u &= (1 - \eta) \int n_u(k, \mu_s, \mu_u) \mu_u(dk). \end{aligned}$$

17.3 RCE objects

An RCE is

$$\{v_i, g_i, n_i, c_i\}_{i=s,u}, \quad r, w_s, w_u, H_s, H_u,$$

where $v_i(k, \mu_s, \mu_u)$ is the value of type i .

17.4 Household problem

For $i \in \{s, u\}$,

$$\begin{aligned} v_i(k, \mu_s, \mu_u) &= \max_{k' \geq 0, n \in [0,1]} U(c, 1 - n) + \beta v_i(k', H_s(\mu_s, \mu_u), H_u(\mu_s, \mu_u)) \\ \text{s.t.} \quad &c = w_i(\mu_s, \mu_u)n + r(\mu_s, \mu_u)k + (1 - \delta)k - k'. \end{aligned}$$

17.5 Firm problem and prices

$$(K^f, N_s^f, N_u^f) \in \arg \max_{K, N_s, N_u \geq 0} \{F(K, N_s, N_u) - rK - w_s N_s - w_u N_u\}.$$

Interior conditions are

$$r = F_K, \quad w_s = F_{N_s}, \quad w_u = F_{N_u}.$$

17.6 Do prices depend only on average capital?

Prices depend on the inputs (K, N_s, N_u) . If labor supplies are endogenous and depend on type-specific capital distributions, then average capital alone is not enough because it does not determine N_s and N_u . If labor supplies are fixed and type masses are fixed, prices can be functions of aggregate capital only. The exam answer must condition on this logic.

18 PS4 Template 3: Tradable versus Nontradable Home Production

18.1 Environment

There are two goods. Good 1 is produced by a representative firm with $F(K, N)$, is tradable, and is the investment good. Good 2 is produced at home using household labor h and idiosyncratic productivity θ :

$$y_2 = \theta h.$$

Households have one unit of time:

$$n + h \leq 1.$$

They value $U(c_1, c_2)$.

The source wording combines capital accumulation with a phrase saying households cannot save/borrow. On an exam, handle the exact wording as follows: if saving is disallowed, drop k' and treat the household problem as static within each period; if capital accumulation is allowed, include k' as a policy and k in the state.

18.2 Tradable good 2

Let p be the price of good 2 in units of good 1. If good 2 is tradable, the household can sell home output. A static within-period budget is

$$c_1 + pc_2 + k' \leq wn + p\theta h + rk + (1 - \delta)k + \Pi, \quad n + h \leq 1.$$

Conditional on total wealth, the consumption choice satisfies

$$\frac{U_2(c_1, c_2)}{U_1(c_1, c_2)} = p.$$

The labor allocation maximizes labor income:

$$\max_{n, h \geq 0, n+h \leq 1} wn + p\theta h.$$

Hence

$$\begin{cases} n = 1, h = 0, & \text{if } w > p\theta, \\ n = 0, h = 1, & \text{if } w < p\theta, \\ \text{any mix,} & \text{if } w = p\theta. \end{cases}$$

Economic intuition

When good 2 is tradable, home labor is just another way to earn market income. The household compares the market wage w to the market value of home production $p\theta$.

18.3 Nontradable good 2

If good 2 is nontradable, it must be consumed by the household that produces it:

$$c_2 = \theta h.$$

The static problem becomes

$$\begin{aligned} \max_{c_1, h} \quad & U(c_1, \theta h) \\ \text{s.t.} \quad & c_1 + k' \leq w(1 - h) + rk + (1 - \delta)k + \Pi, \\ & h \in [0, 1]. \end{aligned}$$

If interior and saving is fixed or separable from the labor split,

$$U_2(c_1, \theta h)\theta = U_1(c_1, \theta h)w,$$

or

$$\frac{U_2(c_1, c_2)}{U_1(c_1, c_2)} = \frac{w}{\theta}.$$

18.4 Efficiency of nontradability

Economist X often claims: “The tradable equilibrium is Pareto efficient, but the nontradable equilibrium is not.” The correct response is: not necessarily. If nontradability is a real feasibility constraint also faced by the planner, the relevant planner problem imposes $c_{2i} = \theta_i h_i$ for each household. The household’s FOC coincides with the constrained planner FOC unless there is another missing market, externality, or distortion. Thus the nontradable equilibrium can be constrained efficient.

Qualifier trap

Do not compare a nontradable equilibrium to a planner who can magically transfer nontradable goods across households. That changes the feasible set. Efficiency must be judged relative to the feasible set in the problem.

19 PS4 Template 4: Sector-Specific Human Capital

19.1 Environment

There are two sectors, A and B . Sector i produces

$$z_i f(H_i),$$

where H_i is efficiency units of labor. Normalize $z_B = 1$ and let $z = z_A$ follow a Markov process $P(z, dz')$. Households have sector-specific human capital (h_A, h_B) and choose one sector in which to work. If they work in sector i ,

$$h'_i = (1 + \varepsilon)h_i, \quad h'_j = (1 - \delta_h)h_j, \quad j \neq i.$$

There are no asset markets. Households own firms equally.

19.2 States

Individual state:

$$x = (h_A, h_B).$$

Aggregate state:

$$S = (\mu, z),$$

where μ is the distribution of households over (h_A, h_B) .

19.3 RCE objects

An RCE is

$$\left(v, d, c, w_A, w_B, H_A^f, H_B^f, \Pi, H\right),$$

where $d(h_A, h_B, \mu, z) \in \{A, B\}$ is the sector choice and $H(\mu, z, z')$ is the distribution law.

19.4 Household problem

Taking wages, dividends, and the aggregate law as given,

$$\begin{aligned} v(h_A, h_B, \mu, z) &= \max_{d \in \{A, B\}} u(c_d) + \beta \int v(h'_A(d), h'_B(d), H(\mu, z), z') P(z, dz') \\ \text{s.t. } c_A &= w_A(\mu, z)h_A + \Pi(\mu, z), \\ c_B &= w_B(\mu, z)h_B + \Pi(\mu, z). \end{aligned}$$

The laws for human capital are

$$\begin{aligned} (h'_A(A), h'_B(A)) &= ((1 + \varepsilon)h_A, (1 - \delta_h)h_B), \\ (h'_A(B), h'_B(B)) &= ((1 - \delta_h)h_A, (1 + \varepsilon)h_B). \end{aligned}$$

19.5 Firm problems

For sector A ,

$$H_A^f(\mu, z) \in \arg \max_{H \geq 0} \{z f(H) - w_A(\mu, z)H\}.$$

For sector B ,

$$H_B^f(\mu, z) \in \arg \max_{H \geq 0} \{f(H) - w_B(\mu, z)H\}.$$

Interior conditions are

$$w_A = z f'(H_A^f), \quad w_B = f'(H_B^f).$$

19.6 Distribution law

For any Borel set $B \subseteq \mathbb{R}_+^2$,

$$H(\mu, z)(B) = \int \mathbf{1}\{(h'_A(d(x, \mu, z)), h'_B(d(x, \mu, z))) \in B\} \mu(dx).$$

Aggregate productivity then evolves separately by $P(z, dz')$.

19.7 Market clearing

$$\begin{aligned} H_A^f(\mu, z) &= \int h_A \mathbf{1}\{d(h_A, h_B, \mu, z) = A\} \mu(dh_A, dh_B), \\ H_B^f(\mu, z) &= \int h_B \mathbf{1}\{d(h_A, h_B, \mu, z) = B\} \mu(dh_A, dh_B), \\ \int c(h_A, h_B, \mu, z) \mu(dh_A, dh_B) &= z f(H_A^f) + f(H_B^f). \end{aligned}$$

20 PS4 Template 5: Heterogeneous Firms with Sector Switching Costs

20.1 Environment

There are two goods, A and B , and a continuum of firms. A firm's productivity in sector A is $z \in \{\bar{z}_1, \dots, \bar{z}_S\}$, iid over time and across firms with probabilities π_s . Productivity in sector B is normalized to one. A firm chooses a sector each period. If it switches relative to last period, it pays cost $\psi \geq 0$. Technology in sector i is

$$z_i f(n), \quad z_A = z, \quad z_B = 1.$$

Households are identical, do not save, own firms equally, supply one unit of labor, and have utility $u(c_A, c_B)$.

20.2 States

Firm individual state:

$$x^f = (j_{-1}, z),$$

where $j_{-1} \in \{A, B\}$ is last period's sector. Aggregate state:

$$\mu \in \mathcal{P}(\{A, B\} \times Z),$$

the distribution of firms over previous sectors and current productivity.

20.3 Prices

Normalize $p_B = 1$ and denote the relative price of good A by $p_A = p$. Let w be the wage in units of good B . The revenue from choosing sector j is $p_j z_j f(n)$.

20.4 Firm value functions

Because switching costs depend on the previous sector, firms are dynamic. Given μ , define

$$\begin{aligned} V(j_{-1}, z, \mu) = & \max_{j \in \{A, B\}, n \geq 0} p_j(\mu) z_j f(n) - w(\mu)n - \psi \mathbf{1}\{j \neq j_{-1}\} \\ & + \beta \sum_{z' \in Z} V(j, z', H(\mu)) \pi(z'). \end{aligned}$$

Let $d^f(j_{-1}, z, \mu)$ be the optimal sector and $n^f(j_{-1}, z, \mu)$ the optimal labor demand.

20.5 Household problem

The household has no dynamic choice. It solves

$$\begin{aligned} \max_{c_A, c_B \geq 0} \quad & u(c_A, c_B) \\ \text{s.t.} \quad & p(\mu)c_A + c_B \leq w(\mu) + \Pi(\mu), \end{aligned}$$

where aggregate profits are

$$\Pi(\mu) = \int \left[p_{d^f} z_{d^f} f(n^f) - w n^f - \psi \mathbf{1}\{d^f \neq j_{-1}\} \right] \mu(dj_{-1}, dz).$$

20.6 Distribution law

For $j \in \{A, B\}$ and $z_s \in Z$,

$$H(\mu)(j, z_s) = \pi_s \int \mathbf{1}\{d^f(j_{-1}, z, \mu) = j\} \mu(dj_{-1}, dz).$$

This formula is often the highest-value part of the answer.

20.7 Market clearing

Labor market:

$$1 = \int n^f(j_{-1}, z, \mu) \mu(dj_{-1}, dz).$$

Goods markets:

$$\begin{aligned} c_A(\mu) &= \int \mathbf{1}\{d^f = A\} z f(n^f) \mu(dj_{-1}, dz), \\ c_B(\mu) &= \int \mathbf{1}\{d^f = B\} f(n^f) \mu(dj_{-1}, dz) - \psi \int \mathbf{1}\{d^f \neq j_{-1}\} \mu(dj_{-1}, dz), \end{aligned}$$

if switching costs are paid in good B . If costs are paid in final utility goods or sector-specific goods, modify the goods-market resource constraint accordingly.

Qualifier trap

The firm's productivity is iid, but the distribution over previous sectors is endogenous because it depends on last period's sector choices. Therefore the aggregate state is not just the productivity distribution.

21 Galina Homework Templates

21.1 Two goods with two agent types and no asset markets

In the sequential-trade two-good problem, type 1 produces the general good x with capital and type 2 produces the specialized good y with capital. The general good can be consumed or invested in either technology; the specialized good must be consumed in the period it is produced.

A sequential-trade CE should list:

- Prices $p_{x,t}$ and $p_{y,t}$, usually normalize $p_{x,t} = 1$ and write $p_t = p_{y,t}$.
- Allocations for each type: $(x_t^i, y_t^i, k_{x,t+1}^i, k_{y,t+1}^i)$ as relevant.
- Individual problems for each type, taking prices as given.
- Market clearing for the general good, specialized good, and capital/investment goods.

Because there are no other markets, do not add bonds or insurance markets.

21.2 Payroll-tax RCE with government spending

Suppose the government spends $G > 0$ each period and finances it by a payroll tax τ :

$$G = \tau w N.$$

If labor is inelastic $N = 1$, the government rule can be written recursively as

$$\tau(S) = \frac{G}{w(S)}.$$

The household's after-tax labor income is $(1 - \tau(S))w(S)n$. Include $\tau(S)$ in the RCE object list or define it as part of the government rule. Feasibility is

$$c + K' = F(K, N) + (1 - \delta)K - G.$$

21.3 Two firm types producing the same final good

Suppose there is a representative household and two equal masses of firms, one with technology $F(k)$ and one with $G(k)$. With no leisure in utility, a compact RCE can use aggregate capital K as the state if households are representative. Objects are

$$v(k, K), \quad g(k, K), \quad r(K), \quad k_F^f(K), \quad k_G^f(K), \quad K'(K).$$

Firm demands solve

$$k_F^f(K) \in \arg \max_{k \geq 0} \{F(k) - r(K)k\}, \quad k_G^f(K) \in \arg \max_{k \geq 0} \{G(k) - r(K)k\}.$$

Capital-market clearing is

$$K = \frac{1}{2}k_F^f(K) + \frac{1}{2}k_G^f(K).$$

If both technologies have infinite marginal product at zero, both firm types use positive capital in equilibrium, so their FOCs hold:

$$F'(k_F^f) = r = G'(k_G^f).$$

22 Past-Qual Templates: Pareto Weights, Externalities, and Entrepreneur RCE

22.1 Pareto weights and initial distribution

A typical complete-markets question asks how a time-0 AD equilibrium depends on the initial distribution of assets. With complete markets and identical homothetic preferences, initial wealth affects Pareto weights and hence consumption shares, but aggregate efficiency conditions may be independent of distribution.

For log utility and a resource constraint $\sum_i c_{it} = C_t$, the Pareto problem

$$\max_{\{c_{it}\}} \sum_i \omega_i \log c_{it} \quad \text{s.t.} \quad \sum_i c_{it} = C_t$$

has FOCs

$$\frac{\omega_i}{c_{it}} = \lambda_t.$$

Thus

$$c_{it} = \frac{\omega_i}{\sum_j \omega_j} C_t.$$

If weights reflect initial wealth, the initial distribution changes individual allocations but not necessarily aggregate capital or output.

22.2 Externality template

Suppose output is

$$y_i = A(K)f(k_i, n_i),$$

where firms take aggregate K as given. In CE, firms set

$$r = A(K)f_k(k_i, n_i).$$

The planner internalizes the effect of aggregate capital on productivity:

$$\text{social marginal product of capital} = A(K)f_k + A'(K)f.$$

The wedge is the missing $A'(K)f$ term. A capital subsidy or investment subsidy can implement the planner allocation if it raises the private return to match the social return.

22.3 Entrepreneur/worker RCE with switching costs

A past-quail style entrepreneur problem has workers who can become entrepreneurs, ideas or productivity shocks, government spending/taxes, and switching costs. The setup is a combination of household/firm dynamic states:

- Individual state includes occupation, assets, idea/productivity, and possibly lagged occupation.
- Value functions are occupation-specific: $V^W(a, z, S)$ and $V^E(a, z, S)$, or one value function with discrete occupation choice.
- Aggregate state includes the distribution over all individual states and government state.
- Government rules specify spending, taxes, transfers, and budget balance.
- Consistency requires the distribution law induced by occupation choices, asset choices, productivity transitions, and government-state transitions.

23 CE versus Planner Wedge Cookbook

23.1 The three-step comparison

Wedge comparison algorithm

1. Derive the CE FOC for the relevant margin.
2. Derive the planner FOC for the same margin under the relevant feasible set.
3. Divide one equation by the other or line them up. The leftover term is the wedge.

23.2 Common wedges

Friction/policy	CE margin	Wedge relative to planner
Capital income tax	$u'_t = \beta u'_{t+1}[1 - \delta + (1 - \tau)F_K]$	Private return below social return.
Changing sales tax	$u'_t = \beta u'_{t+1} \frac{1+s_t}{1+s_{t+1}} R_{t+1}$	Intertemporal relative price wedge.

Friction/policy	CE margin	Wedge relative to planner
Constant sales tax without leisure	Same factor on every date's consumption price	No real intertemporal wedge.
Sales tax with leisure	$U_\ell/U_c = w/(1+s)$	Labor-leisure wedge.
Labor income tax	$U_\ell/U_c = (1-\tau_n)w$	Labor supply too low relative to planner.
Externality $A(K)$	Firm uses private MPK	Planner includes derivative of $A(K)$.
Missing insurance	Household Euler has individual marginal utilities and constraints	Not full-risk-sharing efficient; may be constrained efficient.
Borrowing constraint	Euler inequality	Constrained households have high marginal utility today.
Nontradable good	Household faces own transformation $c_2 = \theta h$	No wedge if planner faces same nontradability.
Switching cost	Private firm accounts for own cost	Potentially efficient if no externality; dynamic state required.

24 Closed-Form Catalog

24.1 Cobb-Douglas prices with inelastic labor

If $Y = K^\alpha N^{1-\alpha}$ and $N = 1$, then

$$r(K) = F_K(K, 1) = \alpha K^{\alpha-1}, \quad w(K) = F_N(K, 1) = (1-\alpha)K^\alpha.$$

This is why PS4 asks whether the rental rate depends only on K .

24.2 Log Pareto sharing

Solve

$$\max_{c_i \geq 0} \sum_i \omega_i \log c_i \quad \text{s.t.} \quad \sum_i c_i = C.$$

The Lagrangian is

$$\mathcal{L} = \sum_i \omega_i \log c_i + \lambda \left(C - \sum_i c_i \right).$$

FOCs give $\omega_i/c_i = \lambda$, so $c_i = \omega_i/\lambda$. Summing over i yields $C = (\sum_i \omega_i)/\lambda$. If $\sum_i \omega_i = 1$,

$$c_i = \omega_i C.$$

24.3 Constant sales tax neutrality without leisure

The CE Euler with sales and capital taxes is

$$\frac{u'(c_t)}{1+s_t} = \beta \frac{u'(c_{t+1})}{1+s_{t+1}} [1-\delta + (1-\tau_{t+1})F_K(K_{t+1})].$$

If $s_t = s_{t+1} = s$ and $\tau_{t+1} = 0$, then

$$u'(c_t) = \beta u'(c_{t+1})[1 - \delta + F_K(K_{t+1})],$$

which is the planner Euler. The constant tax cancels.

24.4 Tradable home-production cutoff

With tradable good 2, the household compares market labor income w to home-production sales revenue $p\theta$. Because income is linear in time allocation,

$$h^*(\theta) = \begin{cases} 1, & p\theta > w, \\ [0, 1], & p\theta = w, \\ 0, & p\theta < w. \end{cases}$$

The cutoff is

$$\theta^* = \frac{w}{p}.$$

24.5 Finite stationary distribution

If a finite-state distribution evolves as

$$\lambda' = B\lambda,$$

a stationary distribution solves

$$\lambda^* = B\lambda^*, \quad \mathbf{1}'\lambda^* = 1, \quad \lambda^* \geq 0.$$

In a stationary RCE, B itself often depends on policy functions, which depend on prices. Thus solving the equilibrium usually means guessing prices, solving policies, constructing B , finding λ^* , and checking market clearing.

24.6 Complete-markets aggregation with identical Euler equations

If all agents face complete markets and share the same discount factor, interior allocations satisfy

$$\frac{U_{i,c}(c_{it})}{U_{i,c}(c_{i,t+1})} = \beta R_{t+1}$$

for all i . With homothetic preferences, individual consumption shares are constant over time, so aggregate consumption obeys a representative-agent Euler equation. This is why time-0 complete-market questions often have aggregate laws independent of initial distribution while individual allocations depend on wealth shares.

25 Exam Variations

25.1 Variation 1: Government budget constraints

Always decide whether the government budget holds period by period or in present value. A period budget has the form

$$G_t + B_t = (\text{tax revenue}) + q_t B_{t+1}.$$

If no debt is allowed, set $B_t = B_{t+1} = 0$. If transfers rebate all revenues, include transfers in the household budget and tax revenue in the government budget.

25.2 Variation 2: Missing markets

If households cannot insure idiosyncratic shocks, do not write a complete-markets planner as the CE benchmark unless the question asks for full efficiency. The relevant CE has individual states and a distribution. The relevant planner comparison may be constrained by the same missing markets.

25.3 Variation 3: Aggregate shocks

If an aggregate shock z affects production and follows $P(z, dz')$, include z in the aggregate state:

$$S = (\mu, z).$$

The household Bellman expectation integrates over both idiosyncratic and aggregate shocks:

$$\beta \int \int v(x', H(\mu, z), z') Q(dx' | x, z) P(z, dz').$$

25.4 Variation 4: Dynamic firms

If a firm has an endogenous state, include a firm value function and a distribution over firm states. If there is a representative household but heterogeneous firms, aggregate state can include K and the firm-state distribution μ^f .

25.5 Variation 5: Price normalization

With multiple goods, normalize one price, usually $p_1 = 1$ or $p_B = 1$. Then write all budget constraints and firm profits in units of the numeraire. Do not normalize two independent prices.

26 Common Traps

1. **Forgetting the aggregate law.** RCE requires $S' = \Lambda(S)$ or $\mu' = H(\mu)$, not only individual policies.
2. **Confusing price sufficiency with state sufficiency.** Prices can depend only on K while future K' depends on the distribution.
3. **Using one value function for all agents.** Dynamic households and dynamic firms need separate value functions.
4. **Leaving out dividends/profits.** If households own firms, profits enter household income, even if profits are zero under CRS.
5. **Omitting government budgets.** Any tax or spending process must be backed by a government budget constraint or rule.
6. **Writing a planner with prices.** A planner uses resource constraints and technology, not market prices, except as multipliers after deriving FOCs.
7. **Declaring inefficiency too fast.** Identify the real wedge relative to the relevant feasible set.
8. **Dropping histories in time-0 AD.** Under uncertainty, commodities are date-history contingent.
9. **Forgetting off-equilibrium states in RCE.** RCE functions should be defined for every feasible aggregate state, not only along the equilibrium path.
10. **Mis-timing capital returns.** Saving at t earns rental income at $t + 1$.

27 Past-Qual Practice Map

Year	Approx. problem	Module 5 connection	Practice focus
2016	Q1	Pareto optimal allocations, time-0 AD equilibrium, role of initial distribution.	Pareto weights and AD definition.
2016	Q3	Asset-pricing RCE structure.	Recursive household problem, market clearing.
2016	Q4	Growth with government consumption and capital/labor taxes.	Time-0 equilibrium, implementability, tax wedges.
2017	Q1	Recursive asset-pricing equilibrium.	RCE object list and pricing functions.
2017	Q2	Endogenous growth with government policy/externality.	CE-vs-planner wedge and implementation.
2020	Q1	Planner allocation and Economist A efficiency claim.	Planner setup and wedge logic.
2020	Q2	Search economy stationary RCE with asset market.	SRCE, invariant distributions, market clearing.
2021	Q2	Asset pricing with taxes and inefficient claim.	RCE and tax wedge reasoning.
2023	Q2	Stock/bond recursive equilibrium.	Recursive equilibrium with assets.
2024	Q1	Disaster-risk planner recursion.	Planner Bellman and state choice.
2024	Q3	Entrepreneur/worker RCE with switching, government state, and taxes.	Multiple values, distribution law, government rule.
PS4 2026	Q1–Q5	Current RCE anchor.	Distribution laws, state sufficiency, heterogeneity, switching costs.

28 Exercise Bank

28.1 A. Mechanical drills

A1. Skill: list equilibrium objects. In the representative growth model with leisure, list the objects in a sequential CE and in an RCE.

A2. Skill: government budget. Add government spending G_t , lump-sum taxes T_t , sales tax s_t , and capital-income tax τ_t to the sequential CE. Write the household budget and government budget.

- A3. Skill: time-0 AD.** In a deterministic economy, derive the present-value budget from the sequential budget using state prices Q_t .
- A4. Skill: distribution law.** Let $x = (a, y)$, $a' = g(a, y, \mu)$, and $y' \sim Q(y, dy')$. Write $\mu'(A \times Y)$.
- A5. Skill: stationary equilibrium.** For a finite-state Markov chain with policy-induced transition matrix B , write the stationary distribution conditions.
- A6. Skill: firm problem.** For a CRS firm $F(K, N)$, derive factor prices and explain why profits are zero.

28.2 B. Closed-form and proof drills

- B1. Skill: Pareto sharing.** Solve the static Pareto problem $\max \sum_i \omega_i \log c_i$ subject to $\sum_i c_i = C$.
- B2. Skill: sales-tax neutrality.** Prove that a constant sales tax is not distortionary in the no-leisure taxed growth model when capital taxes are zero.
- B3. Skill: labor wedge.** Add leisure and show why a positive sales tax distorts labor supply.
- B4. Skill: K-only insufficiency.** Give a formal condition under which aggregate K is not a sufficient aggregate state in a heterogeneous-agent RCE.
- B5. Skill: home-production cutoff.** Derive the labor allocation rule when home good 2 is tradable.
- B6. Skill: nontradable constrained efficiency.** Write the constrained planner FOC for nontradable home production and compare it to the household FOC.

28.3 C. Past-qual and PS4 applications

- C1. Skill: PS4 taste shocks.** Define an RCE for the inelastic-labor taste-shock model and state whether the rental rate depends only on K .
- C2. Skill: skilled/unskilled workers.** Define an RCE for the fixed skilled/unskilled economy. Do prices depend only on average capital?
- C3. Skill: sector-specific human capital.** Define the individual and aggregate states and the distribution law.
- C4. Skill: heterogeneous firms.** Define the firm value function and distribution law for the two-sector switching-cost model.
- C5. Skill: past externality problem.** Suppose firms take $A(K)$ as given but the planner internalizes it. Identify the wedge and propose an implementation policy.
- C6. Skill: stationary search RCE.** A continuum of workers has employment states and asset holdings. What objects must a stationary RCE contain?

28.4 D. Trap and Economist-claim questions

- D1. Claim:** “If good 2 is nontradable, equilibrium cannot be Pareto efficient.” Agree or disagree.
- D2. Claim:** “Any positive tax makes the CE allocation inefficient.” Agree or disagree.
- D3. Claim:** “In the taste-shock RCE, since prices depend only on K , the aggregate state can always be K .” Agree or disagree.
- D4. Claim:** “A model with heterogeneous firms never needs household value functions.” Agree or disagree.
- D5. Claim:** “If goods, labor, and capital markets clear, the government budget can be ignored.” Agree or disagree.

D6. Claim: “A stationary RCE means every household stays in the same state forever.” Agree or disagree.

29 Answer Sketches

29.1 A. Mechanical drills

A1. Sequential CE objects: sequences of allocations $\{c_t, n_t, k_{t+1}, K_t, N_t\}$ and prices $\{w_t, r_t, \Pi_t\}$. RCE objects: $v(k, K)$, policies g_k, g_n , price functions $r(K), w(K)$, firm demands k^f, n^f , aggregate law $K'(K)$, consistency and market clearing. Avoid writing $v(k)$ alone.

A2. Household budget:

$$(1 + s_t)c_t + k_{t+1} \leq w_t n_t + (1 - \tau_t)r_t k_t + (1 - \delta)k_t + \Pi_t - T_t.$$

Government budget:

$$G_t = T_t + s_t c_t + \tau_t r_t K_t$$

if no debt and representative household. With debt, add bond terms.

A3. Multiply each sequential budget by Q_t and sum over t . Choose $Q_{t+1}/Q_t = 1/R_{t+1}$ so capital terms telescope. Use the transversality/no-Ponzi condition to drop the limiting capital term.

A4. For Borel sets A and Y ,

$$\mu'(A \times Y) = \int \mathbf{1}\{g(a, y, \mu) \in A\} Q(y, Y) \mu(da, dy).$$

Do not replace this with only the mean asset law.

A5. The stationary distribution solves

$$\lambda^* = B\lambda^*, \quad \mathbf{1}'\lambda^* = 1, \quad \lambda^* \geq 0.$$

If B depends on policies, solve policies first for guessed prices.

A6. Interior CRS firm FOCs are $r = F_K(K, N)$ and $w = F_N(K, N)$. Euler's theorem gives $F = F_K K + F_N N$, so profits $F - rK - wN = 0$.

29.2 B. Closed-form and proof drills

B1. FOC $\omega_i/c_i = \lambda$. Therefore $c_i = \omega_i/\lambda$. Sum to get $\lambda = (\sum_i \omega_i)/C$. If weights sum to one, $c_i = \omega_i C$.

B2. CE Euler:

$$\frac{u'(c_t)}{1 + s_t} = \beta \frac{u'(c_{t+1})}{1 + s_{t+1}} [1 - \delta + F_K(K_{t+1})].$$

If $s_t = s_{t+1}$, the tax terms cancel and the planner Euler remains. Common mistake: forget that a changing sales tax does not cancel.

B3. With leisure,

$$\frac{U_\ell}{U_c} = \frac{w}{1+s}$$

in CE, while the planner has $U_\ell/U_c = w$. A positive s lowers the private reward to working in consumption units.

B4. K is insufficient if there exist μ_1, μ_2 with $\int k d\mu_1 = \int k d\mu_2 = K$ but

$$\int g_k(x, \mu_1) d\mu_1 \neq \int g_k(x, \mu_2) d\mu_2.$$

Then K' is not a function of K alone.

B5. Tradable home production yields income $wn + p\theta h$ with $n + h \leq 1$. Put all time in the higher-return activity: market work if $w > p\theta$, home production if $p\theta > w$, indifference if equality. Cutoff: $\theta^* = w/p$.

B6. Nontradable planner chooses c_{1i} and h_i with $c_{2i} = \theta_i h_i$. FOC:

$$U_{2i}\theta_i = U_{1i}w$$

if labor in good 1 has marginal product/wage w . This equals household FOC, so no inefficiency unless another wedge is present.

29.3 C. Past-qual and PS4 applications

C1. RCE objects are $v(k, \theta, \mu)$, $g(k, \theta, \mu)$, $c(k, \theta, \mu)$, $r(\mu)$, $w(\mu)$, $H(\mu)$. Prices under Cobb-Douglas/inelastic labor depend only on $K = \int k d\mu$: $r = \alpha K^{\alpha-1}$ and $w = (1 - \alpha)K^\alpha$. The distribution law is

$$H(\mu)(A \times B) = \int \mathbf{1}\{g(k, \theta, \mu) \in A\} Q(\theta, B) d\mu.$$

C2. Use type-specific values v_s, v_u and distributions μ_s, μ_u . Prices are $r = F_K$, $w_s = F_{N_s}$, $w_u = F_{N_u}$. They depend on average capital only if N_s, N_u are fixed or determined by average capital; otherwise they depend on type-specific distributions through labor supply.

C3. Individual state is (h_A, h_B) ; aggregate state is (μ, z) . Sector choice $d \in \{A, B\}$ determines next human capital. Distribution law pushes μ through the policy-induced map from (h_A, h_B) to (h'_A, h'_B) .

C4. Firm state is (j_{-1}, z) . Firm value:

$$V(j_{-1}, z, \mu) = \max_{j, n} \{p_j z_j f(n) - wn - \psi \mathbf{1}[j \neq j_{-1}] + \beta \sum_{z'} V(j, z', H(\mu)) \pi(z')\}.$$

Law:

$$H(\mu)(j, z_s) = \pi_s \int \mathbf{1}\{d^f(j_{-1}, z, \mu) = j\} d\mu.$$

C5. CE ignores $A'(K)$. Planner marginal product includes $A'(K)f$. A subsidy to capital/investment can implement the planner if it makes the private after-tax/subsidy return equal to the social marginal product.

C6. A stationary search RCE contains worker value functions by employment/assets/shocks, policy functions, prices or wage rules, an invariant distribution over worker states, government/unemployment insurance rules if present, and asset/labor/goods market clearing.

29.4 D. Trap and Economist-claim questions

D1. Disagree as stated. Nontradability can be a feasibility constraint. If the planner also cannot transfer good 2, CE can be constrained efficient.

D2. Disagree. Lump-sum taxes are nondistortionary. A constant sales tax without leisure creates no intertemporal wedge. The correct test is whether CE and planner FOCs differ.

D3. Disagree in general. Current prices may depend only on K , but future laws may depend on the distribution. Need μ unless aggregation is proven.

D4. Disagree. If households save, search, choose human capital, or choose occupations dynamically, household value functions are needed. Heterogeneous firms may also need firm value functions.

D5. Disagree. The government budget is part of feasibility. Ignoring it can create fake resources or hide transfers/taxes that affect household choices.

D6. Disagree. Stationarity means the cross-sectional distribution is invariant. Individual states can keep changing.

30 Final Study Checklist

Before attempting old quals, make sure you can do the following without notes:

1. Write the five-block CE checklist: objects, household, firm, government, clearing/consistency.
2. Define time-0 commodities and state prices under uncertainty.
3. Convert a sequential budget into a present-value budget.
4. Write $v(k, K)$ and explain why K is included.
5. Write a distribution law in measurable-set form.
6. State the difference between price sufficiency and state sufficiency.
7. Define a stationary RCE and interpret $\mu^* = H(\mu^*)$.
8. Derive the planner Euler and compare it to CE Euler with taxes.
9. Explain why constant sales taxes are neutral without leisure but distort with leisure.
10. Define RCEs for PS4 Q1–Q5 from memory.
11. Identify when firms need value functions.
12. Answer nontradability and tax-efficiency claims with equations, not slogans.

31 Quality Audit

This packet includes an explicit source inventory, module map, canonical equilibrium templates, planner and welfare comparisons, explicit distribution laws, PS4 applications, Galina tax/homework wedges, past-qual practice mapping, common traps, exercises, answer sketches, and final checklist. The main caveat is that some past-qual mappings are approximate because the relevant local TeX files vary in formatting and topic labels; the packet uses them to identify recurring archetypes rather than to reproduce exact exam wording.